

A Federated Learning Approach for Layer 2 Network Intrusion Detection with Explainable AI Across Distributed Network Segments

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Abstract—The rapid expansion of networked environments, the proliferation of IoT devices, and increasing reliance on cloud services have significantly broadened the attack surface of modern networks. While existing Intrusion Detection Systems (IDS) primarily operate at Layer 3 and above, critical attacks such as ARP Spoofing, MAC Flooding, DHCP Starvation, and STP-based exploits occur at the Data Link Layer (Layer 2) and evade traditional IP-based detection mechanisms. This paper proposes a federated learning framework for Layer 2 network intrusion detection that integrates a hybrid CNN-BiLSTM deep learning model with Explainable AI (XAI) techniques. By leveraging the FedProx aggregation algorithm, the system addresses Non-IID data heterogeneity across distributed network segments without transmitting raw traffic data, thereby preserving privacy. SHAP (SHapley Additive exPlanations) is employed to interpret model decisions, enhancing transparency and trust for security analysts. The proposed framework is evaluated against centralized baselines across varied heterogeneity conditions, with performance measured in terms of accuracy, F1-score, false alarm rate, communication overhead, and computational resource consumption. The research targets practical deployment on resource-constrained hardware and is applicable to enterprise networks, IoT infrastructures, and edge computing environments.

Index Terms—Federated Learning, Intrusion Detection System, Layer 2 Security, Explainable AI, SHAP, CNN-BiLSTM, FedProx, Non-IID Data, ARP Spoofing, MAC Flooding

I. INTRODUCTION

The pervasiveness of computer networks and the accelerating adoption of Internet of Things (IoT) and cloud computing have dramatically increased the attack surface that organizations must defend. Intrusion Detection Systems (IDS) serve as a critical component of network security infrastructure, monitoring traffic patterns to identify anomalous behavior and potential threats [2].

Existing IDS solutions predominantly operate at the network layer (Layer 3) and above, focusing on IP-based traffic features. This design creates a fundamental blind spot: numerous severe attacks occur at the Data Link Layer (Layer 2) before traffic ever reaches IP-level security controls such as firewalls. Layer 2 attacks—including ARP Spoofing, MAC Flooding, DHCP Starvation, and Spanning Tree Protocol

(STP) manipulation—operate at the Local Area Network (LAN) infrastructure level, rarely leaving a direct IP footprint, making them particularly difficult to detect with conventional systems [3], [4].

Beyond the layer-specific gap, mainstream deep learning-based IDS solutions are predominantly centralized, requiring the transfer of raw network data to a central server. This approach creates communication bottlenecks in large-scale distributed networks and raises serious data privacy concerns. Furthermore, deep learning models typically function as black boxes, limiting the ability of security analysts to understand and verify the rationale behind detection decisions—a critical requirement in operational security settings [1].

This paper addresses these three intertwined challenges by proposing a unified framework that combines *federated learning* for distributed, privacy-preserving model training, a *CNN-BiLSTM* deep learning architecture for simultaneous spatial and temporal feature extraction, and *SHAP-based Explainable AI* for model interpretability. The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the proposed framework. Section IV describes the research methodology. Section V discusses expected results, and Section VI concludes with future directions.

II. RELATED WORK

A. Intrusion Detection at the Data Link Layer

Most IDS research has concentrated on network- and transport-layer traffic. Verkerken et al. [6] proposed a multi-stage hierarchical IDS that demonstrates the benefit of layer-aware detection but still relies heavily on IP-level features for its classification stages. Gombao [3] introduced BCAST IDS, a system targeting broadcast-based Layer 2 anomalies, yet the approach remains centralized and does not address privacy constraints in multi-segment deployments. Kilincer [4] explored an explainable AI-supported hybrid deep learning approach specifically for Layer 2 intrusion detection, showing that SHAP-based attribution can expose meaningful MAC-level features; however, the work assumes a single centralized

training environment. These studies collectively demonstrate both the importance and the relative immaturity of Layer 2-focused IDS research, particularly in distributed settings.

B. Federated Learning for Intrusion Detection

Federated learning (FL) has gained considerable traction as a privacy-preserving alternative to centralized model training. Belenguer et al. [1] provide a comprehensive review of FL applications in IDS, noting that the majority of federated IDS proposals target IoT traffic or upper-layer network flows. Their survey identifies Non-IID data distribution as the primary challenge degrading FL model performance in heterogeneous environments and highlights FedProx as a promising mitigation strategy. Chinnasamy et al. [2] survey deep learning-driven network IDS methods and confirm that centralized architectures dominate the literature, with federated variants remaining comparatively underexplored.

C. Explainable AI in Network Security

Explainability is increasingly recognized as essential for operational deployment of machine learning-based security tools. Satilmis et al. [5] review host-based IDS literature and note that the lack of interpretability is a consistent barrier to adoption in enterprise environments. SHAP has emerged as a preferred explanation technique due to its theoretical grounding in cooperative game theory and its ability to produce both global feature importance rankings and local per-prediction attributions. Despite this, its application within federated IDS—where explanations must be generated locally without centralizing data—remains largely unexplored.

D. Research Gap

The survey of related work reveals that while federated IDS [1], Layer 2 detection [3], [4], and XAI for security [5] have each been studied in isolation, no prior work simultaneously addresses all three in the context of Data Link Layer attack detection under Non-IID conditions. This research directly targets that gap.

III. PROPOSED FRAMEWORK

A. Framework Overview

The proposed system is a three-tier federated architecture comprising local monitoring agents, a federated aggregation server, and a distributed explainability module, as illustrated conceptually in Fig. ?? . The key design principle is that raw Layer 2 traffic never leaves its originating network segment; only model parameters are communicated.

Figure 1 illustrates the proposed framework.

B. Tier 1 – Local Monitoring Agents

Each agent is deployed at a distinct LAN segment and performs two functions: feature extraction and local model training. At the feature extraction stage, raw Layer 2 frames are captured and transformed into structured feature vectors including MAC address entropy, ARP request rate, broadcast-to-unicast ratio, DHCP transaction counts, and STP Bridge

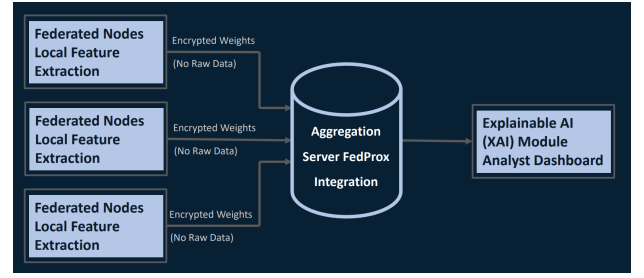


Fig. 1. Conceptual architecture of the proposed Federated Layer 2 IDS.

Protocol Data Unit (BPDU) frequency anomalies. These features are specifically chosen to characterize the behavioral signatures of targeted Layer 2 attacks without requiring IP-layer information.

The local model is a CNN-BiLSTM hybrid. Convolutional layers extract spatial feature patterns within fixed-length traffic windows, while the Bidirectional LSTM layers model temporal dependencies and sequential evolution of traffic behavior. This combination is particularly effective for the bursty, stateful nature of Layer 2 attacks such as MAC Flooding and ARP Spoofing.

C. Tier 2 – Federated Aggregation with FedProx

After each local training round, agents transmit only their updated model weights to the aggregation server. The server applies the FedProx algorithm, which extends the standard FedAvg approach by adding a proximal regularization term $\mu \|w - w^t\|^2$ to each local objective function. This term penalizes excessive deviation of local models from the current global model, improving convergence stability when client data distributions are heterogeneous (Non-IID). The resulting global model is redistributed to all agents for the subsequent round.

D. Tier 3 – Distributed SHAP Explainability

Upon global model convergence, each local agent computes SHAP values for a representative set of detection events using its private traffic data. SHAP assigns each feature a Shapley value quantifying its marginal contribution to a given prediction. Security analysts can inspect these values to identify which Layer 2 features (e.g., abnormal ARP rate, MAC entropy spike) most strongly influenced a specific alert, enabling informed triage and response without exposing raw traffic to a central party.

E. Target Attack Classes

The framework is designed to detect the following Layer 2 attack types:

- **ARP Spoofing:** Gratuitous ARP replies used to poison neighbor caches and enable man-in-the-middle interception.
- **MAC Flooding:** High-volume forged MAC address frames intended to exhaust switch CAM tables and force hub-mode broadcasting.

- **DHCP Starvation / Spoofing:** Exhaustion of the DHCP address pool or injection of rogue DHCP responses to redirect client traffic.
- **STP Manipulation:** Injection of superior BPDU frames to force topology changes and redirect traffic through attacker-controlled bridges.

IV. RESEARCH METHODOLOGY

A. Research Variables

Independent variables are the design decisions that constitute the experimental factors of this study, summarized in Table I.

TABLE I
INDEPENDENT VARIABLES

Variable	Description
DL Architecture	CNN-BiLSTM vs. CNN-only vs. LSTM-only baselines
Training Paradigm	Federated (FedProx, FedAvg) vs. centralized
Heterogeneity	Non-IID degree α (Dirichlet distribution)
XAI Method	SHAP feature attribution

Dependent variables are the evaluation metrics used to assess system performance, listed in Table II.

TABLE II
DEPENDENT VARIABLES (EVALUATION METRICS)

Metric	Description
Accuracy	Overall correct classification rate
Precision	True positives among all predicted positives
Recall	Fraction of actual attacks correctly detected
F1-Score	Harmonic mean of precision and recall
False Alarm Rate	Benign traffic incorrectly flagged as attack
Latency	Per-sample inference time (ms)
CPU / RAM Usage	Resource consumption during training and inference
Comm. Overhead	Model parameter bytes exchanged per federation round

B. Dataset and Simulation Environment

Experiments will be conducted using publicly available Layer 2-enriched network traffic datasets, supplemented by traffic generated in a controlled virtual network environment built with GNS3 and Mininet. Attack traffic will be synthesized using Scapy and standard penetration testing tools to produce labeled ground truth for ARP Spoofing, MAC Flooding, DHCP Starvation, and STP manipulation scenarios.

To simulate a federated environment, captured traffic will be partitioned across virtual clients using a Dirichlet distribution with varying concentration parameter α to control the degree of Non-IID heterogeneity. Low α values produce highly skewed, heterogeneous distributions; high values approach IID conditions.

C. Experimental Design

The study follows a comparative experimental design with five configurations: (1) centralized CNN-BiLSTM baseline, (2) federated FedAvg with IID data, (3) federated FedAvg with Non-IID data, (4) federated FedProx with Non-IID data, and (5) deployment of the best-performing federated model on Raspberry Pi 4 hardware to assess resource feasibility.

Each configuration is evaluated across all dependent variables. Statistical significance of performance differences is assessed using paired t -tests with $p < 0.05$ as the significance threshold.

D. Research Questions

Primary Question: Can the integration of federated learning, deep learning, and Explainable AI improve the detection of Layer 2 network attacks in distributed settings while preserving data privacy?

Sub-questions:

- 1) Can a federated Layer 2 IDS achieve detection accuracy comparable to a centralized counterpart?
- 2) How does Non-IID data heterogeneity affect convergence, and how effectively does FedProx mitigate this compared to FedAvg?
- 3) Which Layer 2 features are most influential in model decisions according to SHAP analysis?
- 4) What communication and computational overhead does federated training introduce relative to centralized training?
- 5) Is the system deployable on resource-constrained hardware while maintaining acceptable performance?

V. EXPECTED RESULTS AND DISCUSSION

A. Detection Performance

The CNN-BiLSTM architecture is expected to outperform single-architecture baselines due to its dual capacity for spatial pattern recognition and temporal sequence modeling. In centralized settings, F1-scores above 0.95 are anticipated for well-represented attack classes such as MAC Flooding and ARP Spoofing, consistent with results reported by Kilincer [4] for Layer 2-focused deep learning models.

Under federated training with IID data, performance is expected to closely match the centralized baseline, validating that the federated paradigm itself does not impose a significant accuracy penalty when data distributions are uniform. Under Non-IID conditions with FedAvg, a measurable degradation in convergence speed and final accuracy is anticipated, particularly for minority attack classes. FedProx is expected to recover a significant portion of this degradation by constraining local model drift, consistent with findings in the federated IDS literature [1].

B. Privacy and Communication Trade-offs

By design, the proposed framework eliminates the transmission of raw traffic data. Communication overhead will be quantified in terms of total parameter bytes exchanged per global round. FedProx introduces a modest computational cost

per local iteration due to the proximal term; this is expected to be offset by faster convergence and fewer required global rounds under Non-IID conditions.

C. Explainability Insights

SHAP analysis is expected to reveal that features related to ARP request rate and MAC address entropy are the dominant contributors to ARP Spoofing and MAC Flooding detections, respectively. These attributions will provide actionable guidance for security analysts and serve as a validation mechanism for model correctness—if SHAP highlights features that are semantically inconsistent with known attack signatures, it would indicate spurious correlations requiring model refinement.

D. Edge Deployment Feasibility

Deployment on Raspberry Pi 4 hardware is expected to demonstrate that the local agent component is viable on constrained devices, with inference latency remaining below practical thresholds for near-real-time detection. Training on such devices will require optimization (e.g., quantization or pruning) and is acknowledged as a direction for future work.

E. Broader Impact

The results of this research will have direct implications for SOC teams, Internet Service Providers, financial institutions, and IoT network operators. The federated, privacy-preserving design aligns with GDPR and similar regulatory frameworks, lowering the barrier to adoption in multi-organizational environments. The open publication of the experimental framework and synthetic dataset generation scripts will facilitate reproducibility and further research in this underexplored area.

VI. CONCLUSION AND FUTURE WORK

This paper has presented a research proposal for a federated, explainable Layer 2 network intrusion detection system. The proposal addresses three critical and interrelated limitations of current IDS solutions: the systematic neglect of Data Link Layer attacks, the privacy and scalability constraints of centralized deep learning architectures, and the opacity of black-box model decision-making.

The proposed framework integrates a CNN-BiLSTM hybrid model for Layer 2 feature extraction, FedProx-based federated aggregation to handle Non-IID data heterogeneity across distributed network segments, and SHAP-based Explainable AI to provide interpretable, per-prediction feature attributions to security analysts. Together, these components form a principled and practically deployable response to each identified limitation.

The experimental design covers centralized baselines, IID and Non-IID federated conditions, and resource-constrained edge hardware validation, ensuring that conclusions are both scientifically rigorous and operationally relevant.

Future work will pursue four extensions. First, differential privacy mechanisms (e.g., Gaussian noise injection on model

gradients) will be investigated to provide formal privacy guarantees beyond the data-locality property of federated learning. Second, cross-domain federated scenarios—where agents represent different organizational types such as enterprise, healthcare, and industrial IoT—will be explored to assess generalization across heterogeneous administrative contexts. Third, continual learning strategies will be developed to enable the global model to adapt to emerging Layer 2 attack variants without full retraining. Fourth, the framework will be extended to support anomaly-based detection of zero-day Layer 2 attacks, which represent the most challenging and operationally significant threat scenario.

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